MSG No. 54.350 P_Bcca [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{3}{4}, 0, 0\right]$	`[1,4,10,11]`
2	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[2,3,9,12]`
3	$\left[\frac{1}{4}, 0, 0\right]$	`[5,8,14,15]`
4	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[6,7,13,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[1,4,10,11]`
2	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,9,12]`
3	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[5,8,14,15]`
4	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[6,7,13,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `..2/m'`

No.	position	mapping
1	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[1,4,13,16]`
2	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[2,3,14,15]`
3	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[5,8,9,12]`
4	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[6,7,10,11]`

Table 4: Wyckoff site: `4d`, site symmetry: `..2/m'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[1,4,13,16]`
2	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[2,3,14,15]`
3	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[5,8,9,12]`
4	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[6,7,10,11]`

Table 5: Wyckoff site: `4e`, site symmetry: `2'/m'..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5,10,14]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,6,9,13]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[3,7,12,16]`
4	$\left[\frac{1}{2}, 0, 0\right]$	`[4,8,11,15]`

Table 6: Wyckoff site: `4f`, site symmetry: `2'/m'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,5,10,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,6,9,13]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,7,12,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[4,8,11,15]`

Table 7: Wyckoff site: `4g`, site symmetry: `m'2m'`

No.	position	mapping
1	$\left[0,\ y,\ \frac{3}{4}\right]$	`[1,3,14,16]`
2	$\left[\frac{1}{2},\ -y,\ \frac{3}{4}\right]$	`[2,4,13,15]`
3	$\left[0,\ -y,\ \frac{1}{4}\right]$	`[5,7,10,12]`
4	$\left[\frac{1}{2},\ y,\ \frac{1}{4}\right]$	`[6,8,9,11]`

Table 8: Wyckoff site: `8h`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{3}{4},\ 0,\ z\right]$	`[1,4]`
2	$\left[\frac{1}{4},\ 0,\ \frac{1}{2}-z\right]$	`[2,3]`
3	$\left[\frac{1}{4},\ 0,\ -z\right]$	`[5,8]`
4	$\left[\frac{3}{4},\ 0,\ z+\frac{1}{2}\right]$	`[6,7]`
5	$\left[\frac{1}{4},\ 0,\ z+\frac{1}{2}\right]$	`[9,12]`
6	$\left[\frac{3}{4},\ 0,\ -z\right]$	`[10,11]`
7	$\left[\frac{3}{4},\ 0,\ \frac{1}{2}-z\right]$	`[13,16]`
8	$\left[\frac{1}{4},\ 0,\ z\right]$	`[14,15]`

Table 9: Wyckoff site: `8i`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{2},\ z\right]$	`[1,4]`
2	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[2,3]`
3	$\left[\frac{1}{4},\ \frac{1}{2},\ -z\right]$	`[5,8]`
4	$\left[\frac{3}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[6,7]`
5	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,12]`
6	$\left[\frac{3}{4},\ \frac{1}{2},\ -z\right]$	`[10,11]`
7	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[13,16]`
8	$\left[\frac{1}{4},\ \frac{1}{2},\ z\right]$	`[14,15]`

Table 10: Wyckoff site: `8j`, site symmetry: `2'..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,10]`
2	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,9]`

continued ...

Table 10

No.	position	mapping
3	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[3,12]
4	$\left[\frac{1}{2}-x,\ 0,\ 0\right]$	[4,11]
5	$[-x,\ 0,\ 0]$	[5,14]
6	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[6,13]
7	$\left[x,\ 0,\ \frac{1}{2}\right]$	[7,16]
8	$\left[x+\frac{1}{2},\ 0,\ 0\right]$	[8,15]

Table 11: Wyckoff site: 8k, site symmetry: 2'..

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	[1,10]
2	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[2,9]
3	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[3,12]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[4,11]
5	$\left[-x,\ \frac{1}{2},\ 0\right]$	[5,14]
6	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[6,13]
7	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[7,16]
8	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[8,15]

Table 12: Wyckoff site: 8l, site symmetry: .2'.

No.	position	mapping
1	$\left[\frac{3}{4},\ y,\ 0\right]$	[1,11]
2	$\left[\frac{1}{4},\ -y,\ \frac{1}{2}\right]$	[2,12]
3	$\left[\frac{1}{4},\ y,\ \frac{1}{2}\right]$	[3,9]
4	$\left[\frac{3}{4},\ -y,\ 0\right]$	[4,10]
5	$\left[\frac{1}{4},\ -y,\ 0\right]$	[5,15]
6	$\left[\frac{3}{4},\ y,\ \frac{1}{2}\right]$	[6,16]
7	$\left[\frac{3}{4},\ -y,\ \frac{1}{2}\right]$	[7,13]
8	$\left[\frac{1}{4},\ y,\ 0\right]$	[8,14]

Table 13: Wyckoff site: 8m, site symmetry: ..m'

No.	position	mapping
1	$\left[x,\ y,\ \frac{3}{4}\right]$	[1,16]
2	$\left[x+\frac{1}{2},\ -y,\ \frac{3}{4}\right]$	[2,15]
3	$\left[-x,\ y,\ \frac{3}{4}\right]$	[3,14]
4	$\left[\frac{1}{2}-x,\ -y,\ \frac{3}{4}\right]$	[4,13]
5	$\left[-x,\ -y,\ \frac{1}{4}\right]$	[5,12]
6	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{4}\right]$	[6,11]
7	$\left[x,\ -y,\ \frac{1}{4}\right]$	[7,10]
8	$\left[x+\frac{1}{2},\ y,\ \frac{1}{4}\right]$	[8,9]

Table 14: Wyckoff site: 8n, site symmetry: m’..

No.	position	mapping
1	$[0,\ y,\ z]$	[1,14]
2	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	[2,13]
3	$\left[0,\ y,\ \frac{1}{2}-z\right]$	[3,16]
4	$\left[\frac{1}{2},\ -y,\ z\right]$	[4,15]
5	$[0,\ -y,\ -z]$	[5,10]
6	$\left[\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[6,9]
7	$\left[0,\ -y,\ z+\frac{1}{2}\right]$	[7,12]
8	$\left[\frac{1}{2},\ y,\ -z\right]$	[8,11]

Table 15: Wyckoff site: 16o, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	[2]
3	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	[3]
4	$\left[\frac{1}{2}-x,\ -y,\ z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$\left[\frac{1}{2}-x,\ y,\ z+\frac{1}{2}\right]$	[6]
7	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	[7]
8	$\left[x+\frac{1}{2},\ y,\ -z\right]$	[8]
9	$\left[x+\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[9]
10	$[x,\ -y,\ -z]$	[10]
11	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	[11]
12	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x,\ -y,\ \frac{1}{2}-z\right]$	[13]
14	$[-x,\ y,\ z]$	[14]
15	$\left[x+\frac{1}{2},\ -y,\ z\right]$	[15]
16	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[16]